

2.3 LFP 预处理流程

$$x_i^{\text{filt}}(t) = \text{Notch}_{49-51\text{ Hz}}(\text{Bandpass}_{1-200\text{ Hz}}(\text{Resample}_{1000\text{ Hz}}(x_i(t))))$$

$$\bar{x}(t) = \frac{1}{N} \sum_{j=1}^N x_j^{\text{filt}}(t)$$

$$y_i(t) = x_i^{\text{filt}}(t) - \bar{x}(t)$$

$$e_{i,k}(\tau) = y_i(t_k + \tau), \quad \tau \in [-0.5, 2.0] \text{ s}$$

2.3 伪迹剔除：稳健统计

$$FlatlineCriterion = 5, \quad ChannelCriterion = 0.8$$

$$LineNoiseCriterion = 4, \quad BurstCriterion = 20, \quad WindowCriterion = 0.25$$

$$RMS_{base} = \sqrt{\frac{1}{|B|} \sum_{t \in B} e(t)^2}$$

$$RMS_{resp} = \sqrt{\frac{1}{|R|} \sum_{t \in R} e(t)^2}$$

$$Q = \frac{RMS_{resp}}{RMS_{base}}$$

$$PTP = \max_{t \in R} e(t) - \min_{t \in R} e(t)$$

$$z_{robust} = 0.6745 \cdot \frac{v - \text{median}(v)}{MAD(v)}$$

$$reject_{robust} = I(z_{robust} > 4)$$

2.3 伪迹剔除：振幅与峰-峰值阈值

$$reject_{\text{amp}} = I \left(\max_t |e(t)| \geq 200 \mu\text{V} \right)$$

$$reject_{\text{ptp}} = I \left(\text{PTP}_e \geq \mu_{\text{PTP}} + 3\sigma_{\text{PTP}} \right)$$

$$reject = reject_{\text{robust}} \cup reject_{\text{amp}} \cup reject_{\text{ptp}}$$

2.4 Spike 预处理：瞬时放电率

$$r_u(t) = \sum_{n=1}^{N_u} K_\sigma(t - s_{u,n})$$

$$K_\sigma(t) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{t^2}{2\sigma^2}\right)$$

$$\sigma = 50 \text{ ms}, \quad \Delta t = 10 \text{ ms}$$

3.2.5 时频分析：Morlet 小波与功率

$$\psi_f(t) = \frac{1}{\sqrt{\sigma_f} \sqrt{\pi}} \exp(2\pi i f t) \exp\left(-\frac{t^2}{2\sigma_f^2}\right)$$

$$W_x(t, f) = \int_{-\infty}^{+\infty} x(\tau) \psi_f^*(\tau - t) d\tau$$

$$P(t, f) = |W_x(t, f)|^2$$

$$P_{\text{dB}}(t, f) = 10 \log_{10} \left(\frac{P(t, f)}{\frac{1}{|B|} \sum_{\tau \in B} P(\tau, f)} \right)$$

$$\Delta P_{c_1-c_2}(t, f) = \overline{P}_{c_1, \text{dB}}(t, f) - \overline{P}_{c_2, \text{dB}}(t, f)$$

4.2.5 主观偏好条件与神经指标

$$C(v) = like, \quad R(v) \geq 4$$

$$C(v) = dislike, \quad R(v) \leq 2$$

$$\Delta P_{like-dislike}(t, f) = \bar{P}_{like, dB}(t, f) - \bar{P}_{dislike, dB}(t, f)$$

$$\Delta r_u(t) = \bar{r}_{u, like}(t) - \bar{r}_{u, dislike}(t)$$

$$slope_u = \frac{\bar{r}_u(t_2) - \bar{r}_u(t_1)}{t_2 - t_1}$$

5.2.2 视频事件标记与唤醒评分

$$w_m = [t_m, t_m + 2 \text{ s}], \quad t_m = t_0 + m \cdot 1 \text{ s}$$

$$audio_m = 0.6 \cdot \text{RMS}_m + 0.4 \cdot \text{Flux}_m$$

$$audio_{m,\text{norm}} = 100 \cdot \frac{audio_m - \min(audio)}{\max(audio) - \min(audio)}$$

$$S_m = 0.7 \cdot arousal_m + 0.2 \cdot confidence_m + 0.1 \cdot audio_{m,\text{norm}}$$

$$Z_m = \frac{S_m - \mu_S}{\sigma_S}$$

5.2.3 功能连接：相干性

$$C_{xy}(f) = \frac{|S_{xy}(f)|^2}{S_{xx}(f)S_{yy}(f)}$$

5.2.3 谱 Granger 因果分析

$$\mathbf{X}_t = \sum_{k=1}^p \mathbf{A}_k \mathbf{X}_{t-k} + \boldsymbol{\varepsilon}_t$$

$$\mathbf{S}(f) = \mathbf{H}(f) \boldsymbol{\Sigma} \mathbf{H}^*(f)$$

$$\mathbf{H}(f) = \left(\mathbf{I} - \sum_{k=1}^p \mathbf{A}_k e^{-i2\pi f k} \right)^{-1}$$

$$F_{x \rightarrow y}(f) = \ln \left(\frac{S_{yy}(f)}{S_{yy|x}(f)} \right)$$

$$F_{y \rightarrow x}(f) = \ln \left(\frac{S_{xx}(f)}{S_{xx|y}(f)} \right)$$

$$D(f) = F_{\text{HPC} \rightarrow \text{AMY}}(f) - F_{\text{AMY} \rightarrow \text{HPC}}(f)$$

5.2.4 Spike 分析：PSTH 与标准化响应

$$\begin{aligned}\text{PSTH}_u(b) &= \frac{1}{N_e \Delta t} \sum_{e=1}^{N_e} \sum_n I\left(s_{u,n}^{(e)} \in b\right) \\ z_u(b) &= \frac{\text{PSTH}_u(b) - \mu_{u,B}}{\sigma_{u,B}} \\ \mathbf{z}_u &= [z_u(b_1), z_u(b_2), \dots, z_u(b_M)]\end{aligned}$$